Stability and Collapse of Pollination Networks:

A Network Science Perspective

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Introduction

- Pollination is one of the most important ecological processes in nature.
- Bees and wasps help transfer pollen between flowering plants.
- Nearly 75% of global crops depend on animal pollination.
- Pollinator populations are declining due to climate change, pesticides, and habitat loss.
- Understanding pollination interactions is important for ecosystem stability.

Motivation

- Pollination systems form complex interaction networks.
- Network science provides tools to study these relationships.
- Helps identify important pollinators that support many plant species.
- Understanding network resilience can help prevent ecosystem collapse.
- Combines ecology, data science, and complex network analysis.

Pollination as a Network

- Pollination interactions can be modeled as a network.
- Nodes represent species.
- Edges represent pollination interactions.
- The system is naturally represented as a bipartite network.

Two types of nodes:

- Pollinators (bees, wasps, insects)
- Plant species

Bipartite Network Structure

- Two sets of nodes:
  - Pollinators
  - Plants
- Links exist only between pollinators and plants.
- No links within the same group.
- Helps represent ecological dependencies clearly.

Network Metrics to Study

Key network science measures:

- Degree Centrality
  - identifies pollinators that interact with many plants
- Betweenness Centrality
  - species that connect different ecological groups
- Eigenvector Centrality
  - influential species connected to other important species
- Modularity
  - Community structure within the network

Research Objective

- Identify important pollinator species in the ecosystem.
- Study how pollination networks are structured.
- Analyze the resilience of the network when species disappear.
- Understand the ecological impact of pollinator decline.

Simulation of Pollinator Loss

- Remove pollinator species from the network.
- Two scenarios:
  - Random species removal
  - Targeted removal of highly connected species
- Measure effects on:
  - Network connectivity
  - Plant survival
  - Network fragmentation
- Adaptive Rewiring Simulation
  - Remove a pollinator.
  - Allow remaining pollinators to form new interactions
  - Compare: Analyze which ecosystem is more resilient.
    - Network without rewiring
    - Network with rewiring

Tools and Technologies

- Python
- NetworkX
- Tulip
- Ecological network datasets

Conclusion

- Pollination networks are complex ecological systems.
- Network science helps analyze species interactions.
- Studying resilience can reveal risks of ecosystem collapse.
- The project demonstrates how computational methods can help protect biodiversity.

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Thank You